

SECTION 15930-A

TERMINAL UNITS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 REFERENCES

- A. Adhesives and insulation materials shall have a composite fire and smoke hazard rating maximum 25 for flame spread and 50 for smoke developed per ASTM E 84. Adhesives shall be waterproof.

1.3 SUBMITTALS

- A. Submit shop drawings and product data in accordance with Division 1.
- B. Submit manufacturer's installation instructions.
- C. Submit manufacturer's descriptive literature, operating instructions, and maintenance and repair data.
- D. Include directions for resetting volume regulators in operating instructions.
- E. Submit detailed box schedule. Indicate box number, cfm, allowable cfm range, pressure drop, inlet configuration, and location. Include schedules listing discharge and radiated sound power level for each of second through sixth octave bands and inlet pressures of 1 inch to 4 inches wg.
- F. Submit discharge and casing-radiated octave band sound power data at operating condition, measured in accordance with ARI Standard 880.
- G. Submit terminal unit controller and local network specification to the owner for approval.
- H. Submit control integration proposal to owner. Interface to FMCS (PLC-based) is required and is the responsibility of the contractor.

PART 2 - PRODUCTS

2.1 ACCEPTABLE MANUFACTURERS

- A. Krueger
- B. Titus
- C. Trane
- D. Tuttle & Bailey
- E. Tek-Air/Conner

2.2 PERFORMANCE

- A. Select units based on performance as scheduled.
- B. Terminal Boxes shall not produce noise levels in excess of the criteria specified below.

Auditorium	NC-15
Executive Offices	NC-25
Conference Rooms	NC-25
Library	NC-30
Private Offices	NC-30
Open Plan Offices	NC-35
Laboratories	NC-40
Computer Equipment Rooms	NC-40
Public Circulation	NC-40

2.3 SINGLE DUCT TERMINAL BOXES

- A. Box Type: Provide boxes of the size indicated and scheduled on the Drawings. Boxes shall be single duct variable air volume and constant air volume, as scheduled.
- B. Construction: Minimum 22 gauge zinc coated steel and lined with a minimum of 1-1/2 pcf internal acoustical-thermal insulation. The entire assembly shall meet the requirements of NFPA 90A and UL 181.
- C. Control Parameters: Fully pressure independent variable volume type with factory preset maximum, minimum flow rates, or special set point field adjustable. Refer to Drawings for control sequence required.
- D. Box Operation: Boxes shall control air flow rate based upon thermostat demand from minimum static pressure from 0.10 inches water up to 4 inches water within plus 5 percent of control volume. Box inlets shall operate to

limit volume variations due to inlet duct configurations to a minimal deviation with the flexible inlet duct bent at 90 degrees.

- E. Control Devices: Provide boxes with pneumatic operators, pneumatic flow controllers and relays to perform the control sequence. Factory test control devices and box operation prior to shipment. Maximum air consumption for control sequence indicated shall not exceed 0.035 scfm at 20 psi.
- F. Amplifying Pressure Probe: Senses air flow rate in inlet. Arrange to permit easy field measurement and/or balancing of air flow rate.
- G. Dampers: Provide with 1 piece neoprene lip seal, capable of shutoff with leakage less than 2 percent of nominal box air flow rating at 4 inches SP.
- H. Sound Attenuator: Provide where scheduled to attenuate inlet valve generated noise to a level which will allow obtaining sound levels as specified. Interior insulation shall be the same as provided for boxes.
- I. Heating Coils: Construct with copper tubes, aluminum fins, and a sheet metal frame. Provide coils fully compatible with boxes. Refer to schedule on Drawings for coil capacities.
- J. Integral Access Panels: Required upstream of heating coils.
- K. Protection: Cap openings in terminal units at the factory, and maintain in sealed condition at Job Site until installation.
- L. Coordinate with the owner to determine the locations of terminal unit controllers. The contractor is responsible to connect all terminal unit controllers as a single connection point to interface to FMCS (PLC-based) The controller data tables and addresses should be organized and provided to the owner.

PART 3 - EXECUTION

3.1 INSTALLATION

- A. Provide an air straightener on the inlet side of the box where 4 duct diameters of straight run do not exist.

- B. Provide access doors or locate above easily removable ceiling components.
- C. Support units individually from the structure. Do not support from adjacent ductwork.
- D. Provide wiring conduits for terminal unit controllers.

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